

Near-Field Focusing via Multi-Dimensional Control Based on AI-Assisted Fast Synthesis

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I. INTRODUCTION AND MOTIVATION

In modern electromagnetic (EM) systems, achieving efficient concentration and precise manipulation of EM energy within a constrained space constitutes a pivotal research direction in antenna and field engineering. Near-field focusing (NFF) technology, characterized by its ability to realize localized energy enhancement within a target region, holds significant application value in fields such as biomedical microwave sensing, wireless power transfer (WPT), and non-destructive testing (NDT). Traditional NFF antenna designs predominantly rely on the Conjugate-Phase Method to achieve energy convergence, supplemented by amplitude weighting techniques to suppress side lobes and enhance focusing efficiency. In these conventional methodologies, the polarization state at the focal point is typically not treated as an independent synthesis variable; instead, it is constrained by the element geometry and the symmetry of the excitation. However, recent advancements have enabled active regulation of the polarization states of antenna array elements, thereby introducing polarization orientation as an additional synthesis variable.

Furthermore, as a multi-objective optimization problem, the design of NFF antennas has become increasingly complex alongside the functional diversification of modern wireless systems. The resulting computational overhead of full-wave simulations and iterative optimizations poses a challenge for real-time and rapid reconfiguration. Against this backdrop, this research introduces the element polarization orientation as an additional degree of freedom (DoF). This approach not only further enhances focusing efficiency but also endows fixed arrays with dynamic polarization reconfigurability, enabling a single array to switch polarization states in real-time to meet varying target requirements. This study proposes the construction of an array-level forward surrogate model to achieve rapid prediction of the EM field distribution in the target region. By integrating optimization algorithms, we implement a three-dimensional joint regulation of amplitude, phase, and polarization orientation at the element level. This framework facilitates the efficient synthesis of complex 3D field distributions and the precise reconstruction of target polarization states in the near-field region.

II. RESEARCH OUTLINE

A. Definition of Surrogate Model Input and Output

For an array with fixed element geometry, topology, and operating frequency, the parameters adjustable during the online phase are the excitation and polarization states. Specifically, for an array composed of $N_x \times N_y$ tunable elements, let the

excitation weight and the polarization control parameter of the n -th element be defined as:

$$w_n = a_n e^{j\phi_n}, \quad p_n \quad (1)$$

where a_n and ϕ_n represent the amplitude and phase control quantities, respectively. The input to the surrogate model is defined as:

$$X = \{\mathbf{A}, \mathbf{\Phi}, \mathbf{P}, z\}, \quad (2)$$

where $\mathbf{A}, \mathbf{\Phi}, \mathbf{P} \in \mathbb{R}^{N_x \times N_y}$ represent the distributions of amplitude, phase, and polarization orientation of the array, respectively. The matrix elements correspond with the states of the physical array elements. z denotes the height of the observation plane parallel to the array in the near-field. While the first three terms characterize the control state of the array at a given instant, the normalized z serves as a conditional variable to specify the observation plane to be queried. Explicitly incorporating z into the input allows the surrogate model to predict the near-field distribution at arbitrary heights within a unified framework, thereby avoiding the curse of dimensionality associated with concatenating multiple planes.

The output of the model is defined as the complete complex electric field distribution Y on the given z -plane. To preserve full EM information regarding phase, amplitude, and polarization vectors, and considering that mainstream learning models are still fundamentally formulated in the real-valued domain [1], the output layer is configured with six 2D channels, corresponding to the real and imaginary parts of the three electric field vector components:

$$Y = \{\Re(E_x), \Im(E_x), \Re(E_y), \Im(E_y), \Re(E_z), \Im(E_z)\} \quad (3)$$

Each channel is defined on a discrete grid of the observation plane, where the grid resolution is configurable to match the desired spatial resolution.

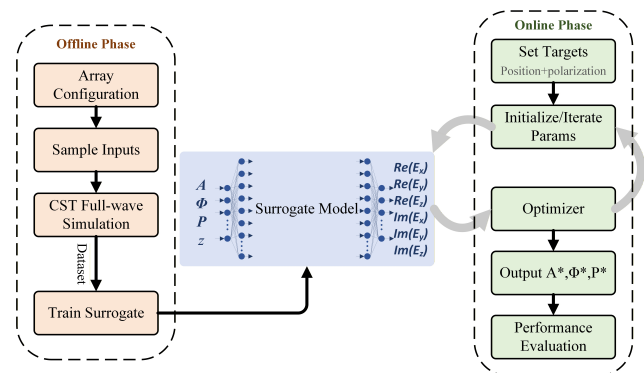


Fig. 1: Workflow of the proposed framework for NFF antenna arrays, consisting of an offline stage and an online stage.

B. Dataset Construction and Model Training

To train the aforementioned surrogate model, a paired dataset covering the control parameter space and the observation height space must be constructed. A 3D sampling space Ω is defined to cover the core radiative near-field region: for an array positioned on the $z = 0$ plane, an appropriate xy observation range is fixed, and the longitudinal height is restricted to the near-field range (e.g., $z \in [0.5\lambda, 10\lambda]$). The dataset is generated using the full-wave EM simulation software CST Microwave Studio. Regarding spatial discretization, the observation region Ω is uniformly partitioned in three dimensions: the sampling points on the xy observation plane correspond to the discrete grid of the output field (divided into $M_x \times M_y$ nodes with a uniform interval), and the z -direction is sampled uniformly with a sub-wavelength step Δz within the near-field range, resulting in M_z observation planes. Consequently, the entire sampling space Ω is divided into $M_x \times M_y \times M_z$ voxel grids, with the simulation output $Y(z)$ at each node serving as the representative field value for that voxel. For the three sets of control parameter matrices ($\mathbf{A}, \Phi, \mathbf{P}$), random sampling is performed within their respective physical constraints to construct training samples covering diverse array excitation states. Each sample is stored as a pair, with the total dataset size large enough to ensure sufficient generalization of the surrogate model across the control parameter and height dimensions. Finally, the effectiveness of the framework will be validated by comparing the surrogate model's prediction accuracy with full-wave simulation results.

C. Optimization Strategy and Control Workflow

1) *Initialization of Control Parameters:* To accelerate convergence and prevent the model from becoming trapped in local optima, the three control parameter matrices must be reasonably initialized before iterative optimization. As noted in [2], the Conjugate-Phase Method remains an effective initial solution for iterative synthesis algorithms; thus, Φ is assigned based on this initial solution. Additionally, the amplitude matrix $\mathbf{A}$ is initialized to a uniform distribution, while the polarization matrix $\mathbf{P}$ is initialized according to the target polarization states of the focal points. In the single-focus scenario, the polarization angles of all elements are set to the angle corresponding to the target co-polarization direction $\hat{\mathbf{e}}_{\text{tar}}$. In the multi-focus scenario, a weighted average of the target polarization angles is used, ensuring that the optimization starting point possesses a coherent superposition basis in the target polarization direction.

2) *Objective Function Definition:* For M target focal points $\{\mathbf{r}_m = (x_m, y_m, z_m)\}_{m=1}^M$, under the total input power constraint $\sum_n a_n^2 = P_{\text{total}}$, the optimization problem is formulated as:

$$\min_{\mathbf{A}, \Phi, \mathbf{P}} \mathcal{L} = \sum_{m=1}^M \lambda_m \mathcal{L}_m \left(\hat{Y}(z_m), \mathbf{r}_m \right), \quad (4)$$

where λ_m is the weight coefficient for the m -th focus. The single-focus loss $\mathcal{L}_m$ consists of two components describing focusing performance and focal shift:

$$\mathcal{L}_m = \mathcal{L}_m^{\text{focus}} + \gamma \mathcal{L}_m^{\text{shift}}, \quad (5)$$

where γ is the penalty weight for focal shift. The focusing loss $\mathcal{L}_m^{\text{focus}}$ simultaneously constrains the field intensity at the focus, polarization matching, and side-lobe levels (SLL):

$$\mathcal{L}_m^{\text{focus}} = - \left| \hat{\mathbf{E}}(\mathbf{r}_m) \cdot \hat{\mathbf{e}}_{\text{tar}}^{(m)} \right|^2 + \alpha \left| \hat{\mathbf{E}}(\mathbf{r}_m) \cdot \hat{\mathbf{e}}_{\perp}^{(m)} \right|^2 + \beta \mathcal{L}_{\text{SLL}} \quad (6)$$

These three terms correspond to co-polarization enhancement, cross-polarization suppression, and SLL penalty, respectively, with α and β as balancing coefficients. The SLL term $\mathcal{L}_{\text{SLL}}$ may require the definition of explicit target main-lobe regions Ω_m^{tar} and background suppression regions Ω_m^{bg} to avoid interference with other focal targets on the same plane.

The focal shift penalty $\mathcal{L}_m^{\text{shift}}$ ensures that the peak field intensity resides near the predefined focal point $\mathbf{r}_m$ rather than drifting to adjacent positions. To this end, a local 3D search neighborhood Ω_m is constructed around the target focus, requiring the field intensity at the target point to be no lower than that at any other sampling point within the neighborhood:

$$\mathcal{L}_m^{\text{shift}} = \max \left(0, \max_{\mathbf{r} \in \Omega_m \setminus \{\mathbf{r}_m\}} \left| \hat{\mathbf{E}}(\mathbf{r}) \right|^2 - \left| \hat{\mathbf{E}}(\mathbf{r}_m) \right|^2 \right), \quad (7)$$

where Ω_m represents the local 3D neighborhood centered at $\mathbf{r}_m$.

3) *Control Workflow:* During the online phase, the user specifies the target focal positions $\{\mathbf{r}_m\}$ and the corresponding polarization states $\{\hat{\mathbf{e}}_{\text{tar}}^{(m)}\}$. The optimizer is warm-started using the aforementioned initialization scheme and employs the surrogate model instead of full-wave simulations for rapid loss function evaluation and iteration as shown in 1. Upon convergence, the optimal control parameters $\{\mathbf{A}^*, \Phi^*, \mathbf{P}^*\}$ are output and dispatched to the array. During and after optimization, the surrogate model can further be utilized for plane-by-plane prediction around z_m to perform post-processing evaluations of key figures of merit including focal gain, -3dB spot size, depth of focus (DoF), and SLL.

III. BUDGET AND CONFERENCE PLAN

The \$2,500 cash award will be used primarily to support the fabrication and experimental validation of the proposed antenna prototypes and other materials required for measurement, tuning, and testing. If necessary, part of the funds may also be used as student stipend support directly related to the research implementation and validation. For the conference, the student plans to attend the 2026 IEEE MTT-S International Wireless Symposium (IWS 2026).

REFERENCES

- [1] R. González Ayestarán, "Support vector regression for the modeling and synthesis of near-field focused antenna arrays," *Electronics*, vol. 8, no. 11, p. 1352, 2019.
- [2] P. Nepa and A. Buffi, "Near-field-focused microwave antennas: Near-field shaping and implementation," *IEEE Antennas and Propagation Magazine*, vol. 59, no. 3, pp. 42–53, 2017.